



UNIVERSIDAD DE GUADALAJARA

1.3. IBM Watson, Kaa, Cisco

Miguel Ángel García Aguilar

Asesor: Karla Ávila Cárdenas

06 de septiembre de 2026

Código:

```
<script setup>
import { computed, onMounted, reactive, ref } from 'vue'

const storageKey = 'biblioteca-crud-libros'
const searchTerm = ref('')
const selectedStatus = ref('Todos')
const showForm = ref(false)
const editingId = ref(null)
const formError = ref('')
const books = ref([])

const emptyForm = () => ({
  title: '',
  author: '',
  year: '',
  genre: 'Novela',
  status: 'Pendiente',
  synopsis: '',
})

const form = reactive(emptyForm())

const filteredBooks = computed(() => {
  const query = searchTerm.value.trim().toLowerCase()

  return books.value.filter((book) => {
    const matchesSearch = !query || [book.title, book.author,
book.genre]
      .some((value) => value.toLowerCase().includes(query))
    const matchesStatus = selectedStatus.value === 'Todos' ||
book.status === selectedStatus.value
    return matchesSearch && matchesStatus
  })
})
})
```

```
const readingCount = computed(() => books.value.filter((book) => book.status === 'Leyendo').length)
const finishedCount = computed(() => books.value.filter((book) => book.status === 'Terminado').length)

function saveBooks() {
  localStorage.setItem(storageKey, JSON.stringify(books.value))
}

function resetForm() {
  Object.assign(form, emptyForm())
  editingId.value = null
  formError.value = ''
}

function openCreateForm() {
  resetForm()
  showForm.value = true
}

function editBook(book) {
  Object.assign(form, book)
  editingId.value = book.id
  formError.value = ''
  showForm.value = true
  window.scrollTo({ top: 0, behavior: 'smooth' })
}

function closeForm() {
  showForm.value = false
  resetForm()
}

function submitForm() {
  formError.value = ''
}
```

```

    if (!form.title.trim() || !form.author.trim() || !form.year
|| !form.synopsis.trim()) {
        formError.value = 'Completa los campos obligatorios para
guardar el libro.'
        return
    }

    const bookData = {
        title: form.title.trim(),
        author: form.author.trim(),
        year: Number(form.year),
        genre: form.genre,
        status: form.status,
        synopsis: form.synopsis.trim(),
    }

    if (editingId.value) {
        const index = books.value.findIndex((book) => book.id ===
editingId.value)
        books.value[index] = { ...books.value[index], ...bookData
    }
    } else {
        books.value.unshift({ id: crypto.randomUUID(),
...bookData })
    }

    saveBooks()
    closeForm()
}

function deleteBook(bookId) {
    const book = books.value.find((item) => item.id === bookId)
    if (!book || !window.confirm(`¿Eliminar "${book.title}" de
tu biblioteca?`)) return

```

```
    books.value = books.value.filter((item) => item.id !==
bookId)
    saveBooks()
}

function statusClass(status) {
    return status.toLowerCase().replace('é', 'e')
}

onMounted(() => {
    const storedBooks = localStorage.getItem(storageKey)
    books.value = storedBooks ? JSON.parse(storedBooks) : [
        {
            id: crypto.randomUUID(),
            title: 'Cien años de soledad',
            author: 'Gabriel García Márquez',
            year: 1967,
            genre: 'Realismo mágico',
            status: 'Terminado',
            synopsis: 'La historia de la familia Buendía y del
pueblo de Macondo a través de siete generaciones.',
        },
        {
            id: crypto.randomUUID(),
            title: 'El problema de los tres cuerpos',
            author: 'Cixin Liu',
            year: 2008,
            genre: 'Ciencia ficción',
            status: 'Leyendo',
            synopsis: 'Una señal extraterrestre conecta el destino
de la humanidad con una civilización al borde del colapso.',
        },
        {
            id: crypto.randomUUID(),
            title: 'Sapiens',
            author: 'Yuval Noah Harari',
            year: 2011,
```

```

        genre: 'Historia',
        status: 'Pendiente',
        synopsis: 'Un recorrido por la evolución de la
humanidad y las ideas que han transformado nuestro mundo.',
    },
]

    if (!storedBooks) saveBooks()
})
</script>

<template>
    <main class="app-shell">
        <header class="topbar">
            <a class="brand" href="#inicio" aria-label="Biblioteca,
inicio">
                <span class="brand-mark">B</span>
                <span>Biblioteca</span>
            </a>
            <span class="topbar-note">Colección personal <span
class="topbar-dot"></span> {{ books.length }} títulos</span>
        </header>

        <section id="inicio" class="hero-section">
            <div class="hero-copy">
                <p class="eyebrow">MI ESTANTERÍA DIGITAL</p>
                <h1>Libros que<br /><em>dejan huella.</em></h1>
                <p class="hero-intro">Organiza tus lecturas, descubre
tus próximos mundos y lleva el pulso de tu colección.</p>
                <button class="primary-button" type="button"
@click="openCreateForm">
                    <span class="button-plus">+</span> Agregar libro
                </button>
            </div>
            <div class="hero-art" aria-hidden="true">
                <div class="sun-disc"></div>
                <div class="book-stack stack-back"></div>

```

```

    <div class="book-stack stack-mid"></div>
    <div class="book-stack stack-front"><span>LECTURAS<br
/>2026</span></div>
    <div class="art-caption">UNA PÁGINA<br />A LA
VEZ</div>
  </div>
</section>

<section class="library-section" aria-
labelledby="library-heading">
  <div class="section-heading">
    <div>
      <p class="eyebrow">TU COLECCIÓN</p>
      <h2 id="library-heading">Todos los libros</h2>
    </div>
    <div class="stats" aria-label="Resumen de lecturas">
      <span><strong>{{ books.length }}</strong>
total</span>
      <span><strong>{{ readingCount }}</strong>
leyendo</span>
      <span><strong>{{ finishedCount }}</strong>
terminados</span>
    </div>
  </div>

  <div class="toolbar">
    <label class="search-box">
      <span class="search-icon">🔍</span>
      <span class="sr-only">Buscar libros</span>
      <input v-model="searchTerm" type="search"
placeholder="Buscar por título, autor o género..." />
    </label>
    <div class="filter-group" aria-label="Filtrar por
estado">
      <button v-for="status in ['Todos', 'Leyendo',
'Pendiente', 'Terminado']" :key="status" type="button"

```

```

:class="{ active: selectedStatus === status }"
@click="selectedStatus = status">
    {{ status }}
</button>
</div>
</div>

<div v-if="filteredBooks.length" class="book-grid">
    <article v-for="book in filteredBooks" :key="book.id"
class="book-card">
        <div class="card-topline">
            <span class="book-index">{{
String(books.indexOf(book) + 1).padStart(2, '0') }}</span>
            <span class="status-badge"
:class="statusClass(book.status)">{{ book.status }}</span>
        </div>
        <div class="book-cover" :class="'cover-
${book.genre.toLowerCase().replaceAll(' ', '-')}'">
            <span class="cover-label">{{ book.genre }}</span>
            <span class="cover-title">{{ book.title }}</span>
            <span class="cover-author">{{ book.author
}}</span>
        </div>
        <div class="book-details">
            <div class="book-heading">
                <h3>{{ book.title }}</h3>
                <span class="book-year">{{ book.year }}</span>
            </div>
            <p class="author">{{ book.author }}</p>
            <p class="synopsis">{{ book.synopsis }}</p>
            <div class="card-actions">
                <button type="button" class="text-button"
@click="editBook(book)">Editar <span>↗</span></button>
                <button type="button" class="delete-button"
aria-label="Eliminar libro"
@click="deleteBook(book.id)">Eliminar</button>
            </div>
        </div>
    </article>
</div>

```



```

        </div>
      </article>
    </div>
    <div v-else class="empty-state">
      <span class="empty-icon">ð</span>
      <h3>No encontramos libros</h3>
      <p>Prueba con otra búsqueda o agrega un nuevo título
a tu colección.</p>
      <button type="button" class="text-button"
@click="searchTerm = ''; selectedStatus = 'Todos'">Limpiar
filtros</button>
    </div>
  </section>

  <footer class="footer">
    <span>Biblioteca personal</span>
    <span>CRUD de libros · Vue.js</span>
  </footer>

  <div v-if="showForm" class="modal-backdrop"
@click.self="closeForm">
    <section class="modal" role="dialog" aria-modal="true"
aria-labelledby="form-title">
      <button type="button" class="close-button" aria-
label="Cerrar formulario" @click="closeForm">x</button>
      <p class="eyebrow">{{ editingId ? 'EDITAR REGISTRO' :
'NUEVO REGISTRO' }}</p>
      <h2 id="form-title">{{ editingId ? 'Actualiza tu
libro' : 'Agrega un libro' }}</h2>
      <p class="modal-intro">Los campos marcados con * son
obligatorios.</p>
      <form @submit.prevent="submitForm">
        <div class="form-grid">
          <label class="field field-wide">Título *<input v-
model="form.title" type="text" placeholder="Ej. Pedro Páramo"
/></label>

```

```

        <label class="field">Autor *<input v-
model="form.author" type="text" placeholder="Nombre del
autor" /></label>
        <label class="field">Año *<input v-
model="form.year" type="number" min="1" max="2100"
placeholder="2026" /></label>
        <label class="field">Género<select v-
model="form.genre"><option>Novela</option><option>Realismo
mágico</option><option>Ciencia
ficción</option><option>Historia</option><option>Ensayo</opti
on><option>Poesía</option><option>Otro</option></select></lab
el>
        <label class="field">Estado<select v-
model="form.status"><option>Pendiente</option><option>Leyendo
</option><option>Terminado</option></select></label>
        <label class="field field-wide">Sinopsis
*<textarea v-model="form.synopsis" rows="4"
placeholder="Escribe una breve
descripción..."></textarea></label>
    </div>
    <p v-if="formError" class="form-error"
role="alert">{{ formError }}</p>
    <div class="form-actions">
        <button type="button" class="secondary-button"
@click="closeForm">Cancelar</button>
        <button type="submit" class="primary-button">{{
editingId ? 'Guardar cambios' : 'Guardar libro' }}</button>
    </div>
</form>
</section>
</div>
</main>
</template>

```